

CSIS 3275 Software Engineering – Practice Lab	
II- Practical	
Instructions: Develop a Java Program solution based on the requirements below.	
General Description: The program will prompt the user to enter client data such as number, name, loan amount, number of years and type of loan. The program will save these data in the database. The program will also have the capability to project the loan schedule based on the number of payments. Other specifications will be specified below.	
You must use H2 as your database for this lab.	
Program Requirements	
1. Create a Spring Boot MVC Project that will have the capability of using H2, JUNIT and MOCKITO. You can add other dependencies (hamcrest, webjars etc.) if you intend to use their capabilities. The output of this step is your updated POM file that contains all the dependencies that you need. You will also create the necessary html files to show your data.	
2. Implement the database in your project by creating table or tables using JPA.	
a. Codes for display, update, delete and edit records	
b. A plain old java object (POJO) of the needed data fields of your database. You will use these fields as your temporary storage for entry and output of data. Derive your POJO files based on the requirements below (Number 4).	3
3. Create your controllers and repositories	5
4. Your index page should show the following fields and interface:	10
Client Loan Records 	
Take note of the following:	
- The page displays the current data that is already inside the table. If your page is new, it will only contain the column headers.	-
- The buttons available are add, edit, delete and amortization table.	-
5. When the user clicks add, the following html file will be shown where the user can add a new entry	5

Add Client Client Number 102 Name Jane Doe Loan Amount 5000 Years 5 Loan Type -- Select -- Save Back	
Note: The Loan Type is a combo box that shows two options: Business or Personal. When the back button is clicked, it goes back to the index page.	
6. Provide the necessary validations needed by the input boxes	3
7. Upon submission, the display html will be shown again with the new entry.	2
Add a message to signify a successful entry	
8. If the user, tries to enter a record of a client where the client number is already existing, the system will not allow this situation, and an error message will be shown.	5

	Client number already exists! Client Number 101 Name John Doe Loan Amount 2000.0 Years 2 Loan Type Business Save Back																			
9. The user can choose a record he or she wants to edit by clicking the edit button of the client The following html file will be displayed.	5																			
	Client Number 101 Name John Doe Loan Amount 30000.0 Years 3 Loan Type Personal Save Back																			
Note: The system should be able to edit correctly any of the fields.																				
The system should also automatically fill out the textboxes of the record chosen for editing.																				
10. The user will edit the record, and the edited record will be displayed after the user clicked the save button. A successful edit message will be displayed	2																			
	Client Loan Records + Add Client <table><tr><th>Client No</th><th>Name</th><th>Loan Amount</th><th>Years</th><th>Loan Type</th><th>Actions</th></tr><tr><td>101</td><td>John Doe</td><td>30000.0</td><td>4</td><td>Personal</td><td> Edit Delete Amortization </td></tr><tr><td>102</td><td>Jane Doe</td><td>2000.0</td><td>2</td><td>Business</td><td> Edit Delete Amortization </td></tr></table>	Client No	Name	Loan Amount	Years	Loan Type	Actions	101	John Doe	30000.0	4	Personal	Edit Delete Amortization	102	Jane Doe	2000.0	2	Business	Edit Delete Amortization	
Client No	Name	Loan Amount	Years	Loan Type	Actions															
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102	Jane Doe	2000.0	2	Business	Edit Delete Amortization															

: The balance on the first month is the loan amount. Follow the following guidelines for computation purposes. - To determine the part of the monthly payment that will go to the interest, multiple the balance with the monthly interest. - To determine the part that will go to the principal subtract the monthly payment with interest. - To determine the new balance, subtract the previous balance with the principal. The above data is not saved in the database. It is computed on the fly using programming logic. - When the user clicks back it will go back to the main page	
 The screenshot displays an 'Amortization Schedule' for a client named Jane Doe. The loan amount is 2000.0, the term is 2 years, and the loan type is Business. The schedule table lists 9 months of payments. Each month, a payment of 91.37 is made. The interest portion decreases from 15.0 in month 1 to 10.3 in month 9. The principal portion increases from 76.37 in month 1 to 81.07 in month 9. The balance starts at 1923.63 and decreases to 1291.69 by month 9. A 'Back' button is visible at the bottom of the schedule.	
12. Your MVC should have an example of implementation of dependency injection.	2
13. You will create a JUNIT and MOCKITO test for the class that implemented the display, update and delete records interface. This will be your system under test (SUT). Please follow the requirements below:	
a. You will use MOCKITO to mock the database dependencies of this Class.	5
c. Create a test for adding, deleting, and displaying records.	9
14. All tests should be passing.	
15. The program should be running correctly.	
16. The program should accurately produce the desired results.	5
17. The program should have proper and ample documentation.	5
18. The program should name its variables and controls mnemonically.	2
Submission Guidelines: 1. Allocate ample time for preparing your files for submission. 2. Check if you are submitting the correct file. There is no chance of file resubmission once the exam ended. 3. Name the file with your student number and first name as part of the file name. Failure to do this will have a one-point deduction on your mark. 4. Zipped the proper files for submission. 5. Test the zipped file if it is opening and running properly.	

6. Submit the file. Note: As this is the final exam, the professor will not have time to tinker with your program. You must ensure your program is running properly. Program deviations such as the following might result to the deduction on the output, program execution and compliance to requirements program criteria. 1. The program is running but the user must press the button twice. 2. The folders needed are not in the project and needs to be transferred manually. 3. Students who use a different IDE runs the risk of the program not running properly.	
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